

Agentic AI – Student Class Notes

5-Day AI & Data Science Employment Readiness Program

DAY 1 – AI/DS, Python & Generative AI

1. Python & Data Science Refresh

- Functions
- Collections – List, Tuple, Set, Dictionary
- File handling
- JSON
- Exception handling
- Python environments
- NumPy
- Pandas
- API requests

2. AI / ML / DL Basics

- Supervised learning – learning from labelled data.
- Unsupervised learning – finding patterns in unlabelled data.
- Training vs inference
- Model evaluation metrics
- Overfitting
- Embeddings
- Basic neural-network concepts

3. LLM Foundations

- Tokens
- Context
- Transformer concept
- Sampling
- System instructions
- Hallucination
- Responsible AI usage

4. Prompt Engineering

- Task, context and constraints
- Examples
- Structured outputs
- Model parameters

- Secure API credentials
- Basic logging

5. Mini Project

Extract role, skills and experience from a job description and convert the information into validated JSON.

DAY 2 – Reliable LLM Applications & RAG

1. Reliable LLM Applications

- Reusable LLM wrappers
- Message roles
- Schemas
- Validation
- Retry mechanisms
- Fallback mechanisms
- Conversation state
- Context control

2. LLM Evaluation & Safety

- Golden questions
- Relevance
- Factuality
- Privacy
- Prompt injection
- Latency
- Token usage
- Cost tracking

3. RAG – Retrieval-Augmented Generation

Basic flow: User Question → Retrieve Information → Give Context to LLM → Generate Answer

- Documents
- Document chunks
- Embeddings
- Vector database/index
- Similarity search
- Retrieved context
- LLM
- Citations

4. Document Ingestion

- Collect documents
- Clean documents
- Split text into chunks
- Add metadata
- Create embeddings

- Store them in a vector index

DAY 3 – RAG in Depth & Agentic AI

1. RAG Pipeline

- Top-k retrieval
- Metadata filters
- Context assembly
- Cited answers
- Safe no-answer behaviour

RAG flow: Documents → Chunking → Embeddings → Vector Index → Retrieval → Context → LLM → Answer

2. RAG Tuning

- Chunk size
- Chunk overlap
- Query rewriting
- Similarity thresholds
- A/B experiments

3. RAG Evaluation

- Retrieval hit rate
- Groundedness
- Citation correctness
- Answer relevance
- Latency
- Failure analysis

4. Agentic AI

- An AI agent works toward a goal using tools and controlled workflows.
- Goal, tools, state, plan-act-observe loop
- Memory boundaries
- Stop conditions

Agent workflow: Goal → Plan → Act → Observe → Decide → Act → Final Answer

5. Tools & Guardrails

- Typed tools
- Allowlists
- Human approval
- Iteration limits
- Tool-error handling

- Logs
- Prompt-injection protection

DAY 4 – Capstone Project

Career Readiness Copilot

The capstone analyses a target job role and student profile, retrieves trusted career information, selects approved tools and creates an evidence-grounded readiness plan.

Main Features

- Student profile input
- Job description input
- Input validation
- RAG knowledge base
- Skill-gap analysis
- Knowledge search
- Interview quiz
- Learning plan
- Agent workflow
- State management
- Visible trace
- Tool error handling
- Human confirmation
- Citations
- Privacy
- Prompt-injection checking

Day 4 Development Flow

- 1 Plan the project
- 2 Design architecture
- 3 Build the knowledge layer
- 4 Implement RAG
- 5 Add tools
- 6 Build the agent controller
- 7 Add state transitions
- 8 Create the interface
- 9 Integrate everything
- 10 Test failure cases

DAY 5 – Deployment & Employment Readiness

1. Hardening

- Validation
- Retry
- Safe fallback
- Citations
- Trace quality
- Secret management
- Latency improvement
- UI improvement

2. Testing

- Retrieval
- Groundedness
- Tool correctness
- Safety
- Latency
- Cost

3. Red-Team Testing

- Prompt injection
- Unsafe requests
- Missing context
- Tool failures
- Too many iterations
- Code clarity

4. Deployment

- Publish/run the application
- Present architecture
- Explain evidence
- Explain limitations
- Explain individual contribution

5. Employment Readiness

- README
- Resume bullet
- 60-second technical pitch
- 30-day learning roadmap
- Post-test

■ QUICK REVISION

Python: Python → Data → API → JSON → Exception Handling

LLM: Prompt → Context → Model → Response

RAG: Documents → Chunks → Embeddings → Vector Search → Context → LLM → Cited Answer

Agent: Goal → Plan → Tool → Observe → State → Next Action → Final Answer

Complete Agentic AI App: User → Interface → Agent → Tools → RAG → LLM → Validation → Response